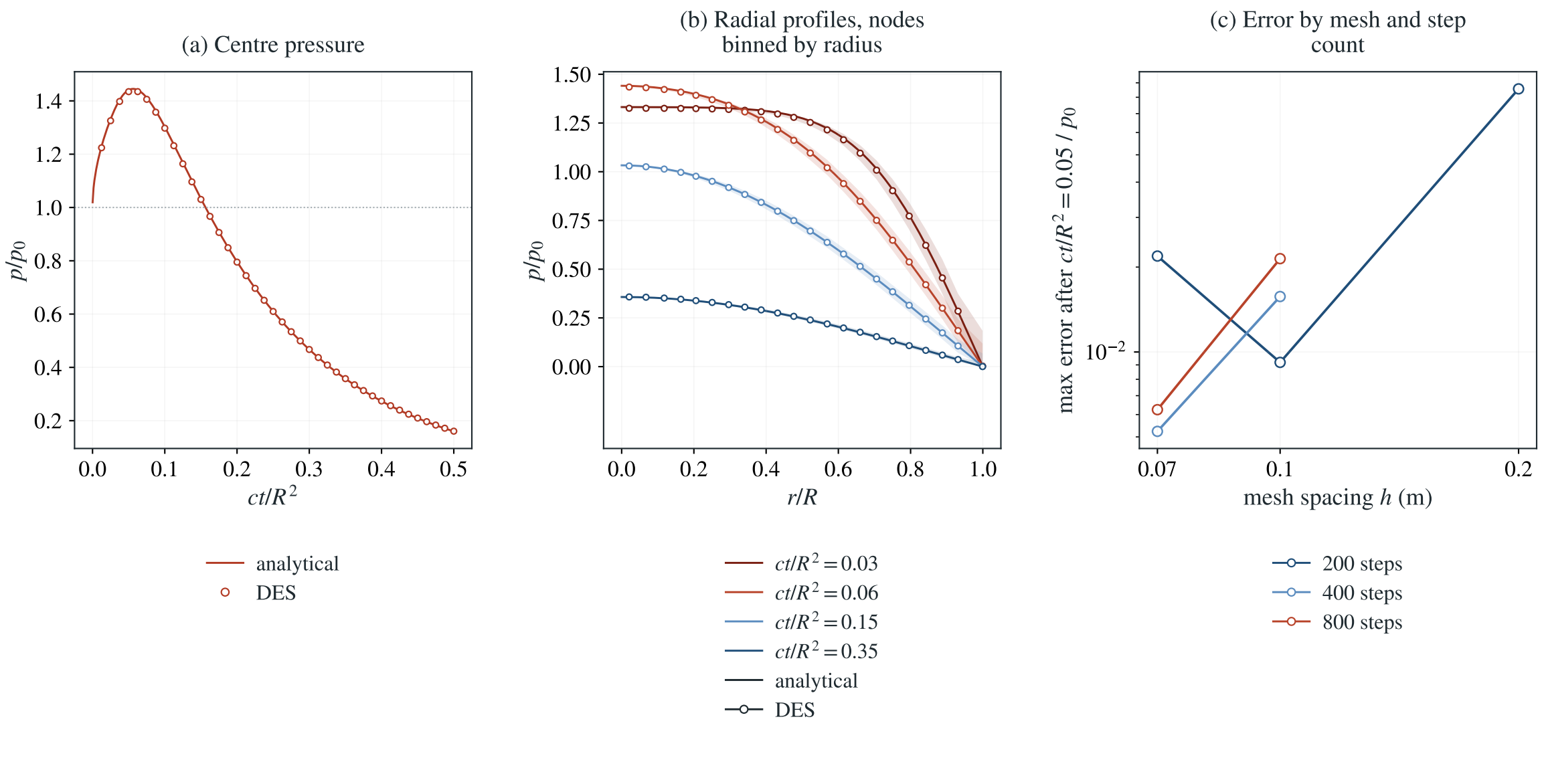

(a) Centre pressure
1.4
1.2
1.0
0.8
0.6
0.4
0.2
p/p_0
0.0
0.1
0.2
0.3
0.4
0.5
ct/R^2
analytical
DES
(b) Radial profiles, nodes binned by radius
1.50
1.25
1.00
0.75
0.50
0.25
0.00
p/p_0
0.0
0.2
0.4
0.6
0.8
1.0
r/R
$ct/R^2 = 0.03$
$ct/R^2 = 0.06$
$ct/R^2 = 0.15$
$ct/R^2 = 0.35$
analytical
DES
(c) Error by mesh and step count
max error after $ct/R^2 = 0.05$ / p_0
10^{-2}
0.07
0.1
0.2
mesh spacing h (m)
200 steps
400 steps
800 steps